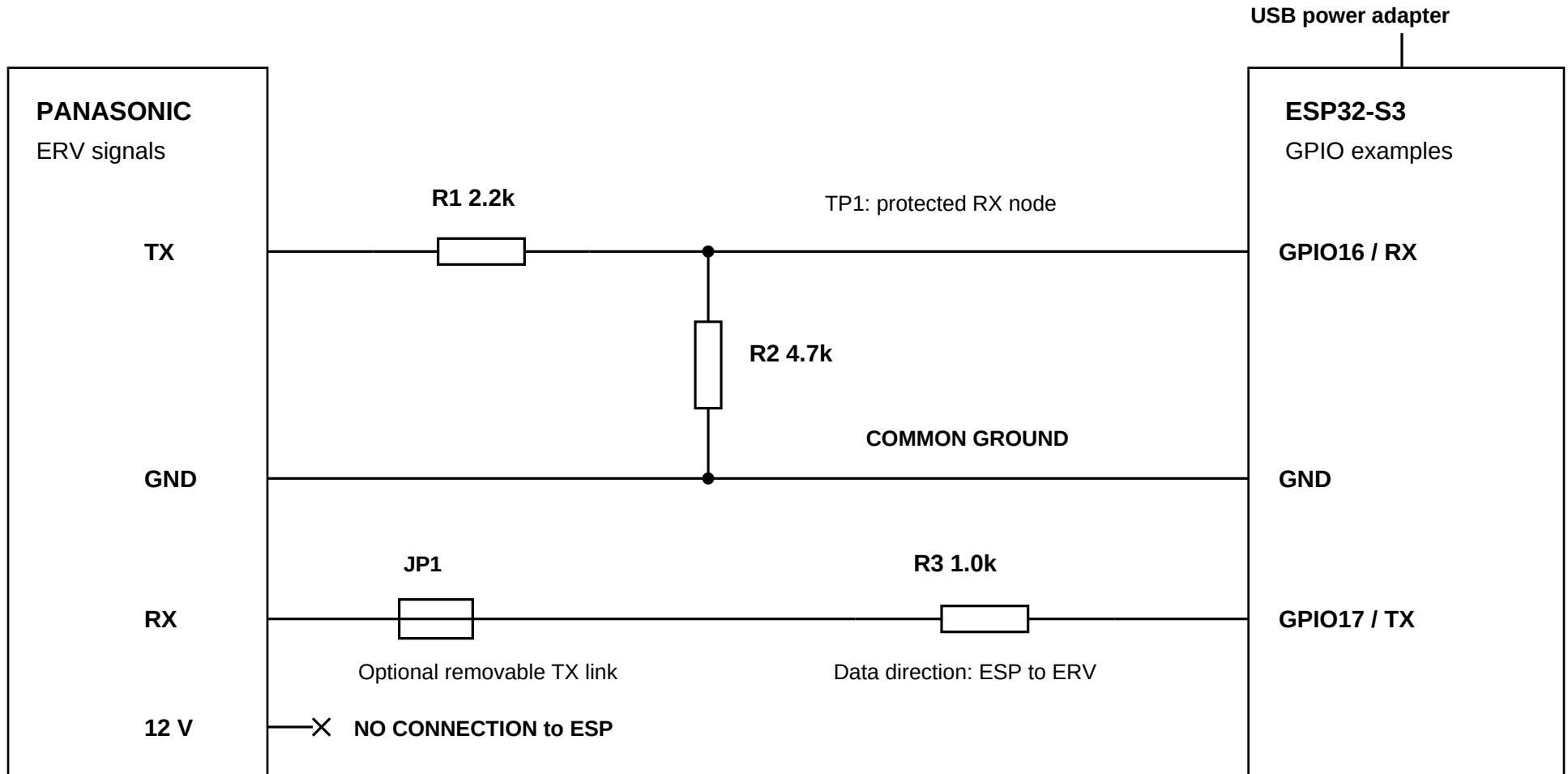


Panasonic FV-16VEC1S / ESP32

Active UART wiring - example GPIO16 RX / GPIO17 TX; pins configurable in ESPHome

ONLY for the verified 4.16-4.48 V ERV TX line. Not suitable for the ~6.7 V controller TX.



OEM controller TX must be disconnected from ERV RX before connecting the ESP transmit path.

R1/R2 divider + nominal 45k internal pull-down: about 2.74-2.95 V at TP1 over the measured range.

Use rx_pull_down: true for this circuit. Without internal pull: about 2.83-3.05 V. Measure your unit.

R1/R2/R3: 1%, 1/4 W. Dots join wires. TX/RX labels are from the ERV perspective.

3.3 V ESP transmit worked on the tested unit. R3 is not a level shifter. Never connect ERV 12 V to ESP.

4800 baud / 8 data / EVEN parity / 1 stop / RX and TX inverted. Verify voltage and direction first.